Distance from a point to a line

1 $d = \dfrac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$

2

a distance: $\dfrac{5}{\sqrt{65}}$ units

b distance: $\dfrac{62\sqrt{5}}{15}$ units

c distance: $\dfrac{28}{5}$ units

d distance: $\dfrac{72}{25}$ units

e distance: $\dfrac{23}{13}$ units

f distance: $2\sqrt{13}$ units

3 $\dfrac{9}{\sqrt{113}}$

4

a $k = 7, k = -27$

b $k = 5, k = 35$

5

a distance: $\sqrt{5}$ units

b $b = \pm\sqrt{319}$

Geometric applications

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a Center: $(0, 0)$

b $r = 0.9$

c $\dfrac{8}{\sqrt{65}}$

d 0, because the perpendicular distance from the centre of the circle to the line is bigger than the radius of the circle.

7

a Center: $(-3, -2)$

b $r = 2$

c $\dfrac{6}{\sqrt{34}}$

d 2, because the perpendicular distance from the centre of the circle to the line is smaller than the radius of the circle.

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a 1 **b** $x - y - 1 = 0$ **c** $\left(-\frac{9}{2}, -\frac{11}{2}\right)$ **d** -1

e Yes **f** isosceles **g** $BD = \frac{9}{2}\sqrt{2}$ **h** $AC = 3\sqrt{2}$

i Area: $\frac{27}{2}$ **j** $(-9, -1)$

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a $3x - 4y + 13 = 0$ **b** Height: 3 units

c Area: 15 units^2

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